Statistical Inference

Probability and Statistics Foundations — Part 3

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Master's in Asset Management
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Outline

1. Random variables and distributions
   How do we describe an uncertain outcome mathematically?

2. Population parameters and moments
   If we know the distribution, which summary numbers matter?

3. **Statistical inference**
   If we *don't* know the distribution, how do we learn it from data?

Answering the Risk–Return Question

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Answer

Using inference.

Expected value $\neq$ Sample mean

Recall from Part 2 — these are two different objects, and only one of them can be seen.

	Expected value	**Sample mean**
Symbol	$\mathbb{E}[X]$	$\bar{X}$
Formula	$\sum_j x_j f(x_j)$	$\frac{1}{n}\sum_{i=1}^{n} x_i$
Built from	the **true** distribution $f(x)$	n **observed** data points
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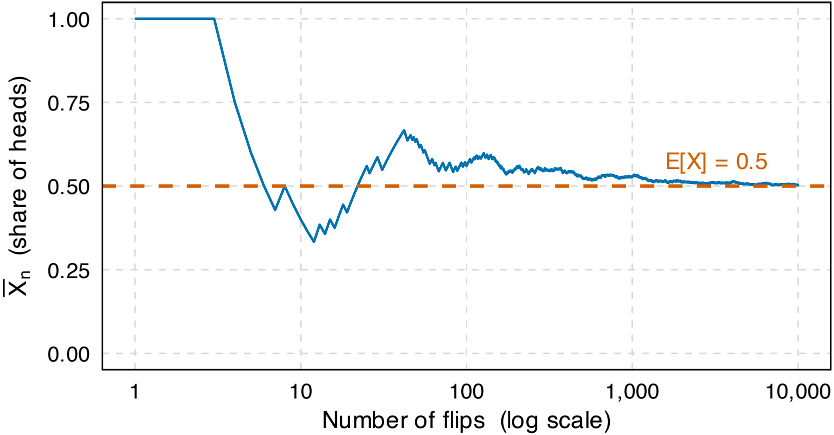
> $\mathbb{E}[X]$ is the **true** value, and it is unknown.
>
> $\bar{X}$ is **known**, because it is built from what we observed — but it is not the thing we wanted.

Remember the coin flip example?



Parameter $\neq$ Statistic

The same split as $\mathbb{E}[X]$ versus $\bar{X}$, now stated for *any* parameter.

	Population parameter	**Sample statistic**
What it is	It describes true distribution	It is a function of the sample
Examples	$\mu,\ \ \sigma^2,\ \ \mathrm{corr}(X, Y)$	$\bar{X},\ \ s^2,\ \ \widehat{\mathrm{corr}}(X, Y)$
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Inference is all about using the **sample statistic** to say something about the **population parameter**.

The Two Fundamental Types of Inference

Estimation

Use the data to put a **number** on the unknown parameter — and say how good that number is.

“My best guess of $\mathbb{E}[R_S]$ is 1% per month.”

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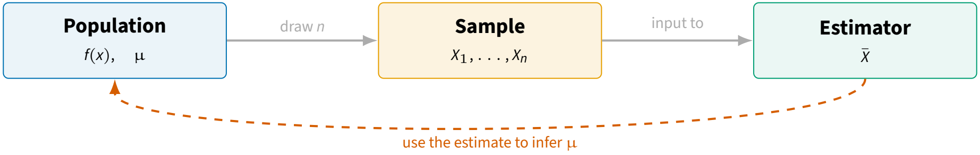
Hypothesis testing

Use the data to rule a claim **out** — to say what the unknown population parameter is *not*.

"Whatever $\mathbb{E}[R_s]$ is, it's very unlikely that it is zero."

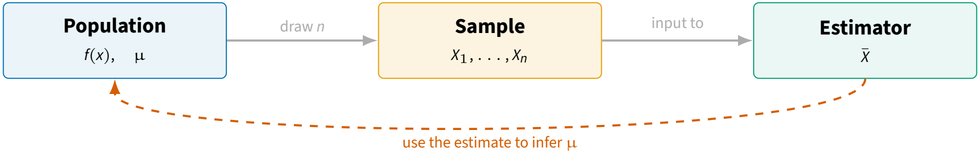
A **negative** approach: it never confirms, it only rejects.

Estimation



Population
$f(x), \quad \mu$
draw n
Sample
$X_1, \ldots, X_n$
input to
Estimator
$\bar{X}$
use the estimate to infer μ

ESTIMATION



Parameter	μ	Fixed, unknown.
Estimator	$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$	A **function** of data. A random variable.
Estimate	$\bar{X} = 0.48\%$	The **number** the function returns on the sample you have.

Sampling from the Population

Before doing any estimation, we need data! The most common way of sampling is the “random” sample.

Random sample

$X_1, \ldots, X_n$ is an **i.i.d. random sample** from a population with pdf $f(x)$ if the draws are **independent** and **identically distributed** — each from that same $f(x)$.

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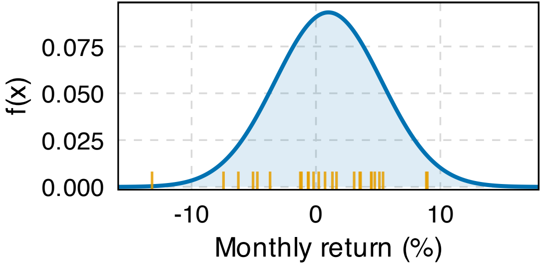

One draw of $n = 25$: the ticks came out of the curve.

All 25 ticks came out of the **same** curve, and the curve does not move between draws — that is the **identically distributed**.

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Independent

Knowing X_1 tells you nothing about X_2: the joint pdf factors,

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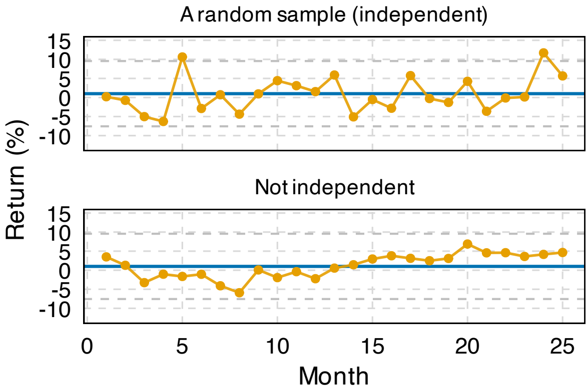

Same μ (blue) and same marginal σ in both; dashed lines are $\mu \pm 2\sigma$.

A New Sample, a New Estimate

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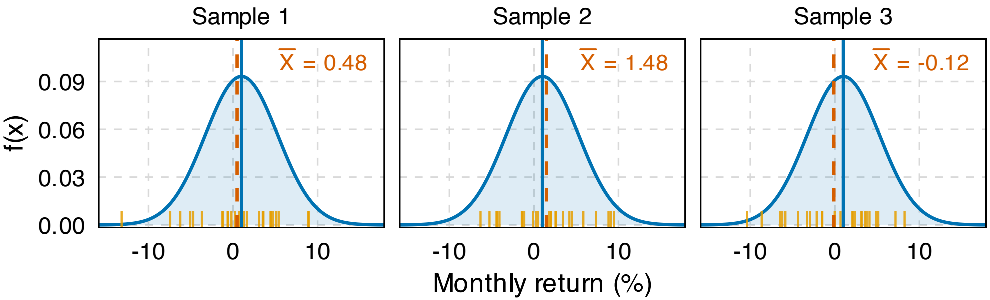

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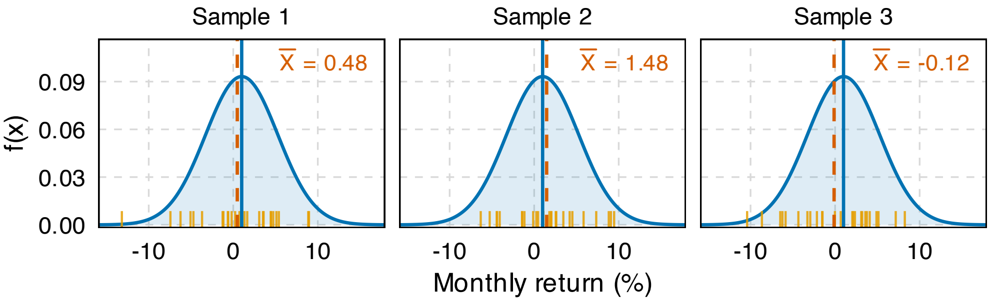

> There is quite a bit of variance in the estimator $\bar{X}$!

The Estimator Has a Distribution of Its Own

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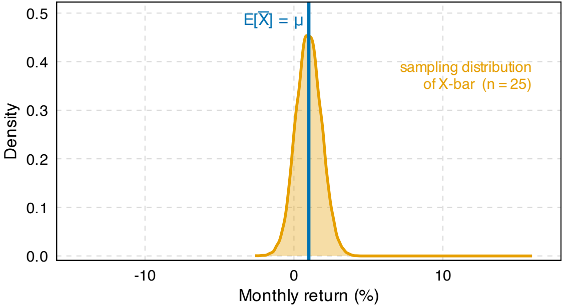

Variance of the estimator

We want $\mathrm{sd}(\bar{X})$ to understand how volatile $\bar{X}$ can be. Using VAR.3, with the covariance term deleted because the draws are assumed **independent**:

$$\begin{aligned}
\mathrm{var}(\bar{X}) &= \mathrm{var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) \\
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Standard error

The standard error is an *estimate* of the standard deviation of the estimator:

$$\mathrm{se}(\bar{X}) = \frac{s}{\sqrt{n}}.$$

Note: Put s in place of the unknown σ!

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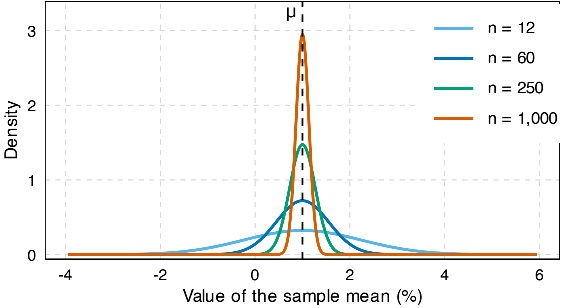

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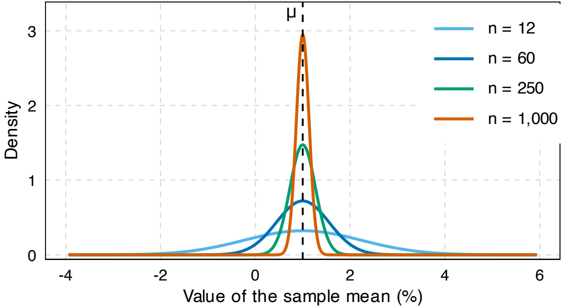

The distribution **narrows**. More data buys **precision**.

But how fast?

Distribution of $\bar{X}$

Assuming i.i.d. sampling, we know two of the three things that describe the distribution of $\bar{X}$:

Where it sits: $\mathbb{E}[\bar{X}] = \mu$

How wide it is: $\mathrm{sd}(\bar{X}) = \sigma/\sqrt{n}$

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Answer

Yes, we can get a Normal! But we need a trick.

The Central Limit Theorem

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Let $X_1, \ldots, X_n$ be i.i.d. with mean μ and *finite* variance σ^2. Then, as n grows,

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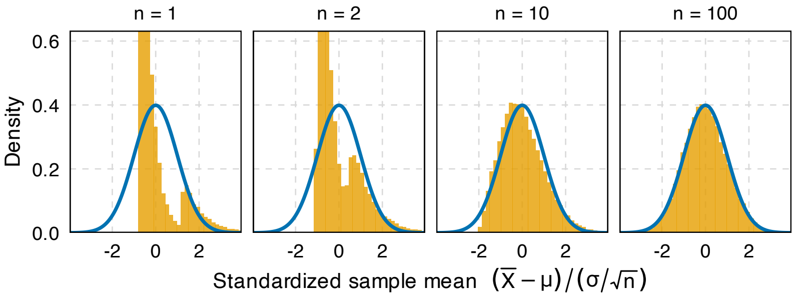
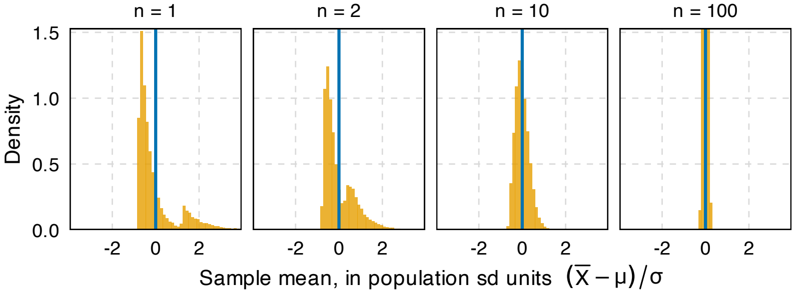
Watching It Happen



A deliberately hideous population: Exp(1.6) with probability 0.85, else $3 +$ Exp(0.8) — right-skewed (skewness $+2.1$), spiky, bimodal. The $n = 1$ and $n = 2$ spikes run off the top; blue curve is the standard normal.

Why Divide by $\sigma/\sqrt{n}$, Not σ?



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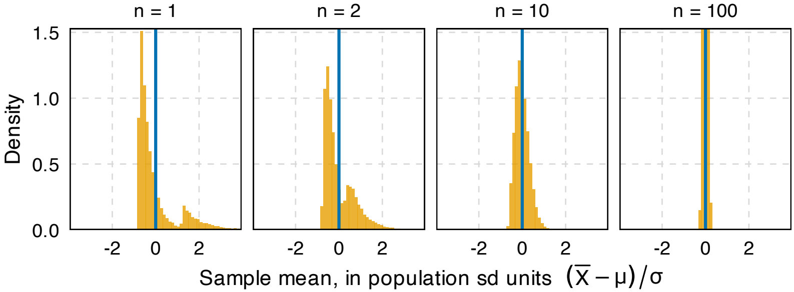

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Without the $\sqrt{n}$ there is no shape to converge to — everything just collapses onto μ.

That collapse is due to the **Law of Large Numbers**.

Estimation produces a number and an uncertainty.

This is more ambitious that saying: "I want to know if the true value is *not* μ."

The two hypotheses

- H_0, the **null** hypothesis. This is presumed true. We try to reject.
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Example: A trial does not find the defendant innocent. It finds them *not guilty*.

Examples of Nulls and Alternatives

The question	H_0**: null**	H_1**: alternative**
Does this fund manager add value?	No added value: $\alpha = 0$	$\alpha > 0$
Do earnings surprises move the price?	No reaction: zero abnormal returns	$\alpha \neq 0$
Does a higher minimum wage affect jobs?	No employment effect: $\beta = 0$	$\beta \neq 0$
Is inflation running at target?	On target: $\mu = 2\%$	$\mu \neq 2\%$

How Do We "Reject" a Null?

STEPS:

1. Assume H_0 is true.

2. Draw the distribution of $\bar{X}$ under H_0.

3. Place realization of $\bar{X}$ on this distribution.
Ask: is this unlikely?

4. Quantify "unlikely."

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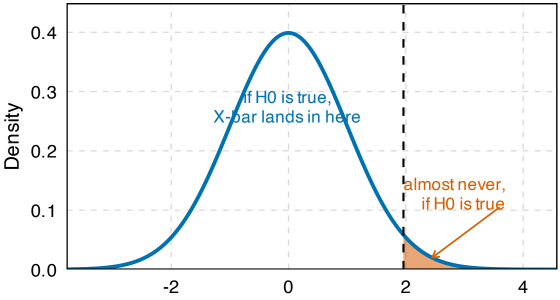

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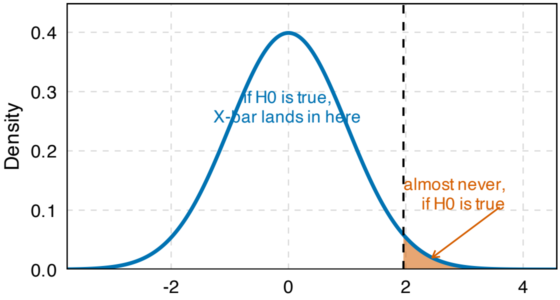
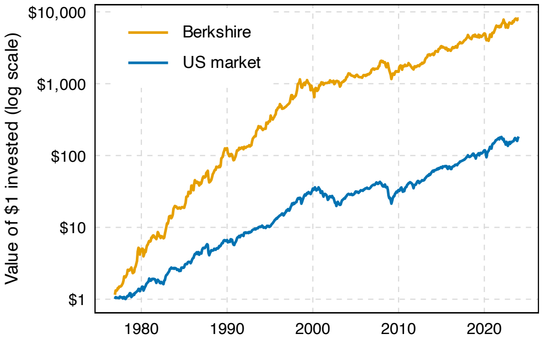


Let's take an example to fix ideas.

Was Buffett Lucky or Skilled?



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Monthly returns, November 1976 – December 2023.

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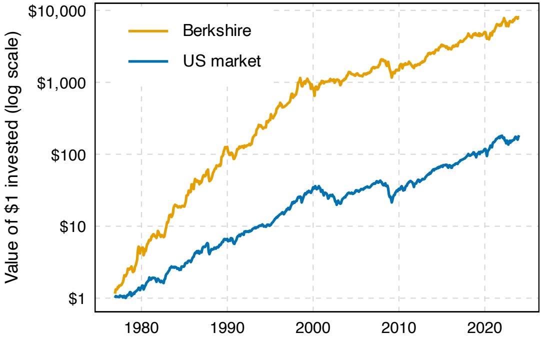

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Question

Is that consistent **skill** or a few **lucky** calls?

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Write Berkshire's return as the market's return plus a wedge:

$$r_t^B = r_t^M + X_t \qquad \Longrightarrow \qquad X_t = r_t^B - r_t^M.$$

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H_0 is the **luck** hypothesis: Berkshire's true edge is zero, the good months and the bad ones cancel, and forty-seven years of outperformance is what a fair coin looks like when you flip it 563 times.

Step 2: The Distribution Under H_0

If H_0 is true, we know **everything** about how $\bar{X}$ behaves:

Centre: $\mathbb{E}[\bar{X}] = \mu_0 = 0$ from H_0

Spread: $sd(\bar{X}) = \sigma/\sqrt{n} = 0.252\%$ assumed

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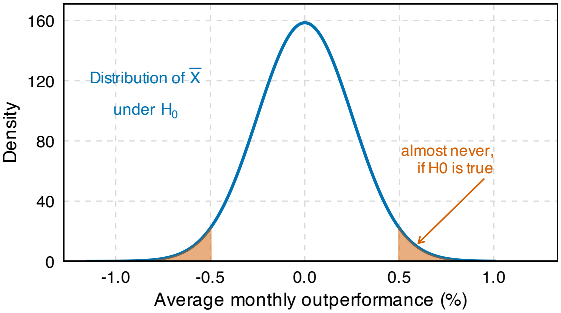

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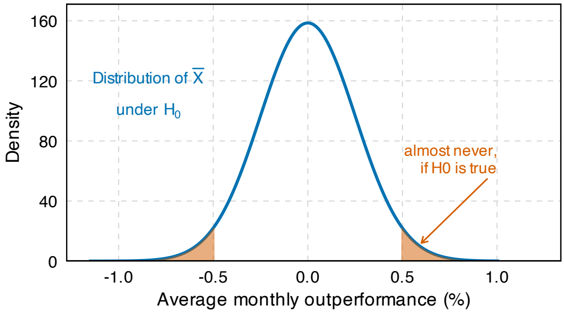
Berkshire delivered $\bar{X} = 0.790\%$ per month.



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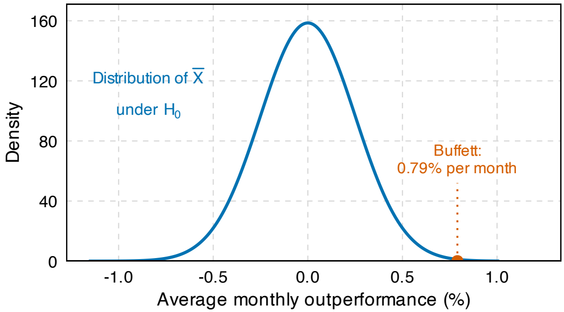
Shape: normal from the CLT, $n = 563$

Berkshire delivered $\bar{X} = 0.790\%$ per month.



Step 3: Placing $\bar{X}$

If H_0 is true, we know **everything** about how $\bar{X}$ behaves:

Centre: $\mathbb{E}[\bar{X}] = \mu_0 = 0$ from H_0

Spread: $\text{sd}(\bar{X}) = \sigma/\sqrt{n} = 0.252\%$ assumed

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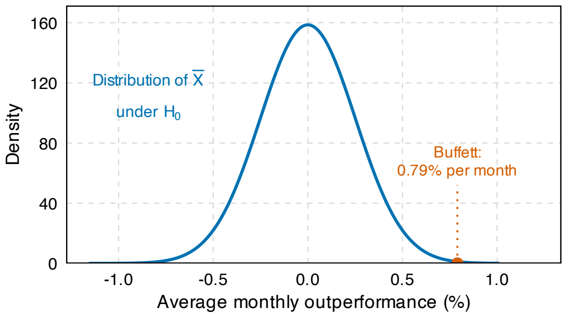

Question

Is $\bar{X} = 0.790\%$ unlikely?

Step 4: Quantify "Unlikely"

Three ways to put a number on how unlikely $\bar{X}$ is given H_0:

1. **Test statistic**
2. *p*-**value**
3. **Confidence interval** Appendix slides

Step 4: Quantify "unlikely" using test statistics

One way to quantify "unlikely" is to measure the distance from the null.

Test statistic

One number saying how far the estimate sits from the null, measured in **its own** standard errors.

The CLT hands us the first version — but it needs the population σ, which we have to assume:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim \mathrm{N}(0, 1).$$

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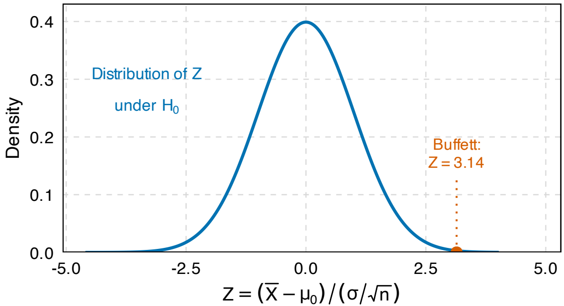

$$z = \frac{0.790 - 0}{0.252} = \mathbf{3.14}$$

Step 5: Rejection rule

Pick the measure

Rejection rule

Test statistic t →

(i) Pick a critical value c.

(ii) Reject H_0 if

$t > c$ or $t < -c$ or $|t| > c$

Question

How to pick c?

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How to pick c?

Option 1: Refer to some external threshold.
ex: c = 1.96
ex: c = 5.7 = Stock market return in **Sep. 1931**, standardized

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How to pick c?

Option 1: Refer to some external threshold.
ex: c = 1.96
ex: c = 5.7 = Stock market return in **Sep. 1931**, standardized

Option 2: Attach a notion of probability. See p-value below.

Step 4: Quantify "Unlikely" using the p-value

p-value

The probability of getting a test statistic **at least as extreme** as the one observed, *assuming H_0 true*.

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Set the critical value to be the test statistic ($c = 3.14$ in our example) and compute the CDF:

$$
\begin{aligned}
&\text{p-value} = P(T > c) \\
\text{OR}\quad &\text{p-value} = P(T < c) \\
\text{OR}\quad &\text{p-value} = P(|T| > |c|)
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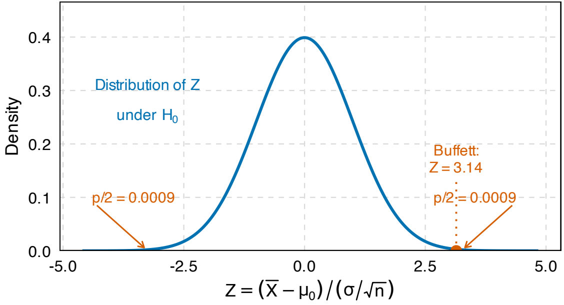
$$



STEP 4: QUANTIFY "UNLIKELY" USING THE p-VALUE

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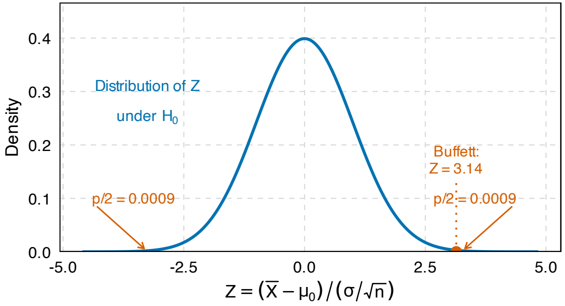

Usually **two-tailed**, because H_1 was $\mu \neq 0$. Interpretation: a result so extreme in either direction is rare.

Step 5: Rejection rule with p-value

Pick the measure

p-value →

Rejection rule

(i) Pick a significance level α.

(ii) Reject H_0 if p-value $< \alpha$

Standard value is $\alpha =$ **0.05**.

Reject H_0 when $p < 0.05$.

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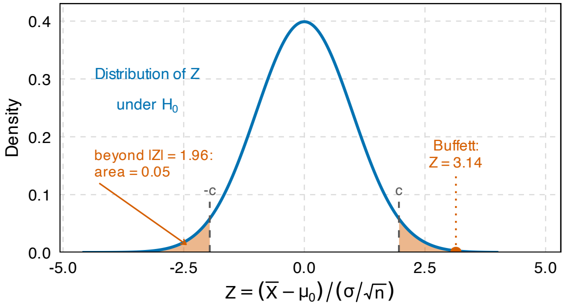
Why 0.05? It says: an observation this large occurs only **1 time in 20 times** when H_0 is true.

More importantly, it's a convention.

Buffett: The Verdict

Route 1 — critical value

$|z| = 3.14 \; > \; 1.96 = c.$ Reject H_0.



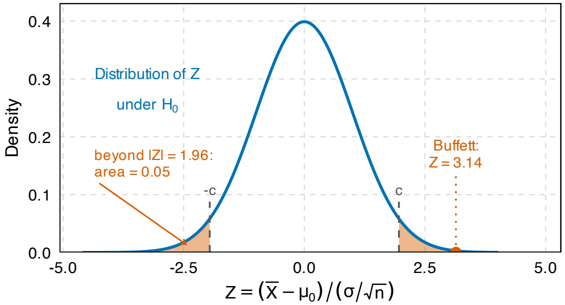
Buffett: The Verdict

Route 1 — critical value

$|z| = 3.14 \; > \; 1.96 = c.$ Reject H_0.

Route 2 — *p*-value

$p = 0.0018 \; < \; 0.05 = \alpha.$ Reject H_0.



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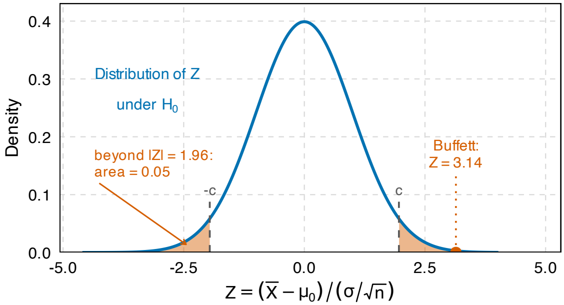
The region corresponds to $|Z| > 1.96$.
Its area is 0.05.
$|Z| > c \iff p < \alpha.$



BUFFETT'S ALPHA

> Warren Buffett's Berkshire Hathaway has realized a Sharpe ratio of 0.79 with significant alpha to traditional risk factors. [...]
>
> Therefore, Buffett's returns appear to be neither luck nor magic but, rather, a reward for leveraging cheap, safe, high-quality stocks.
>
> Decomposing Berkshire's portfolio into publicly traded stocks and wholly owned private companies, we found that the public stocks have performed the best, which suggests that Buffett's returns are more the result of stock selection than of his effect on management.

Frazzini, Kabiller and Pedersen (2018), *Financial Analysts Journal* **74**(4).

Key Takeaways

Parameters vs. statistics	Parameters (μ, σ^2) are fixed and invisible; statistics ($\bar{X}$, s^2) are visible and random.
Inference	Using a **sample statistic** to say something about an unknown **population parameter**.
Estimation	Put a number on the parameter and say how good it is — a **positive** claim about what it *is*.
Estimators	A rule, not a number. Judge it by its sampling distribution: unbiased, efficient, consistent.
Standard error	The dispersion of an estimator.
Hypothesis testing	Rule a claim **out** — a **negative** claim about what the parameter is *not*. It never confirms.
The CLT	For large n, $\bar{X}$ is approximately normal *whatever* the population.
Test statistic	Number saying how far the estimate is from the null
p-value	Probability of getting a test statistic at least as extreme as the one observed, if H_0 true.

Three Parts, One Argument

Part 1

An uncertain outcome is a **random variable**, it gives rise to a distribution.

Part 2

A distribution is summarised by its **moments**.

Part 3

We never observe those moments. We **estimate** them, or we **reject** what they are not.

References

Frazzini, Andrea, David Kabiller, and Lasse Heje Pedersen. 2018. "Buffett's Alpha." *Financial Analysts Journal*, 74(4): 35–55.

Harvey, Campbell R., Yan Liu, and Heqing Zhu. 2016. "…and the Cross-Section of Expected Returns." *Review of Financial Studies*, 29(1): 5–68.

What Makes an Estimator Good

Some estimators are better than others. Here are some important characteristics a good estimator has:

1. Unbiased

The estimator is centred on the population parameter.

2. Efficient

Among estimators that are unbiased, it has the smallest *variance*.

3. Consistent

The estimator's sampling distribution collapses onto the population parameter as n grows large.

Unbiasedness

Efficiency

Consistency

Unbiasedness

Unbiased

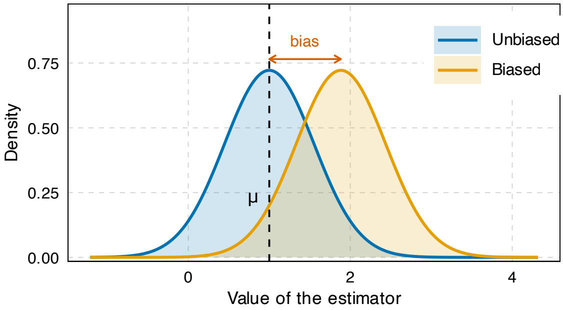
$\hat{\theta}$ is unbiased for θ if

$$\mathbb{E}\left[\hat{\theta}\right] = \theta \quad \text{for every } \theta.$$

For the sample mean this is just E.3 from Part 2:

$$\mathbb{E}[\bar{X}] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \frac{1}{n}n\mu = \mu.$$

Bias is $\mathbb{E}[\hat{\theta}] - \theta$: the gap between where the sampling distribution sits and where the truth is.



Back

Efficiency

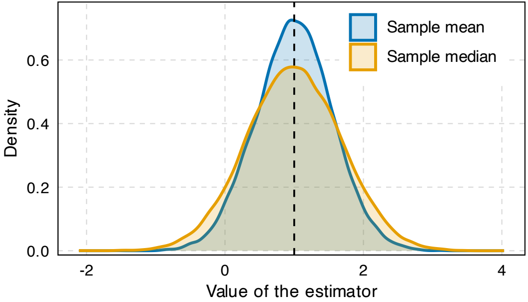
Unbiasedness alone is a low bar: the *first observation* X_1 is an unbiased estimator of μ too, and it is obviously terrible.

Efficiency

Among unbiased estimators, prefer the one with the **smallest variance**.

Example: The sample median as an estimator for the expected value.

- The sample median is unbiased because the population is symmetric.
- But the median's sampling distribution is **wider** than the mean's.



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Consistency

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$$\hat{\theta} \xrightarrow{p} \theta \qquad \text{as } n \to \infty.$$

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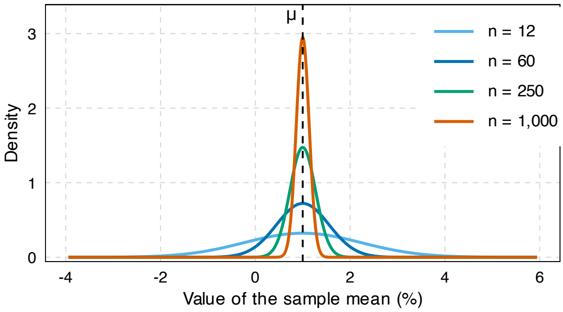
Here it is again, from the other side: not one path over time, but the whole distribution tightening.



One year, five years, twenty years, eighty years of monthly data.

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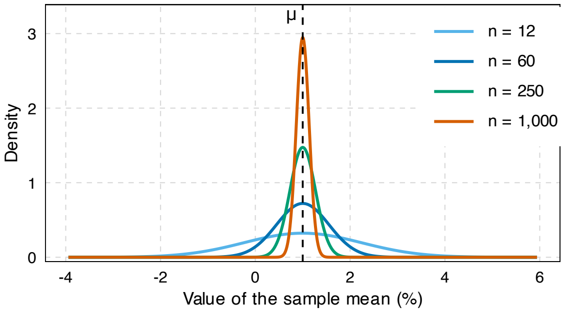

One year, five years, twenty years, eighty years of monthly data.

> The curve does not shift — it **narrows**. Consistency is a statement about spread, not about location.

Back

Step 4: Quantify "unlikely" using confidence intervals

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with $c = 1.96$ for 95%.

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Question

What does this confidence interval measure?

Answer

Across repeated samples, 95% of the intervals built this way contain μ.

Step 5: Rejection rule with confidence interval

Pick the measure

Confidence interval

Rejection rule

(i) Pick a confidence level $1 - \alpha$.

(ii) Reject H_0 if μ_0 lies **outside** $\bar{X} \pm c \cdot \mathrm{se}(\bar{X})$

Standard value is $1 - \alpha =$ **95%**

The same cut a third time: μ_0 outside the interval $\iff |Z| > 1.96 \iff p < 0.05$.

But there are diminishing gains...

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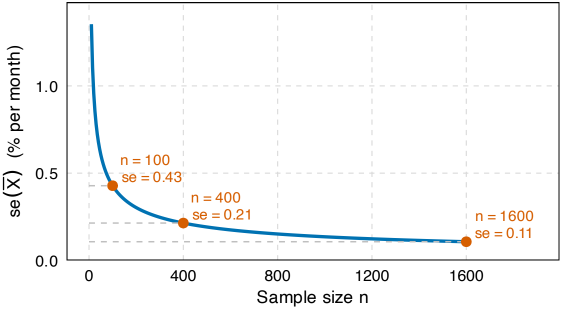

Standard error of the sample mean, with $\sigma = 4.28\%$ per month.

Note that for time series, we cannot even extend the sample size. We currently have 100 years of data, or 1200 months. We cannot $\times 4$ our sample size!

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